EXP. NO: 24

MATLAB SIMULATION OF PASSBAND TRANSMISSION WITH BER ANALYSIS UNDER MULTIPATH FADING CHANNELS

Objectives

1. To simulate a digital passband communication system using MATLAB and study transmission through a multipath fading channel.
2. To generate a digitally modulated passband waveform, transmit it through a multipath fading channel, and recover the received data.
3. To evaluate system performance using Bit Error Rate (BER) under different signal-to-noise ratio (SNR) conditions.
4. To analyze the effect of multipath propagation, delay spread, fading, and additive noise on the received signal and BER performance.
5. To plot and interpret BER versus SNR characteristics and compare the simulated results with the expected communication-system behavior.

CODE

```
clc;
clear;
close all;

% Parameters
N = 1000;
SNR = 10;

% Generate random bits
bits = randi([0 1], N, 1);

% BPSK Modulation
tx = 2*bits - 1;

% Multipath Channel
h = [1 0.5 0.3];
fade = filter(h, 1, tx);

% Add AWGN Noise
noisePower = mean(fade.^2) / (10^(SNR/10));
noise = sqrt(noisePower) * randn(N,1);
rx = fade + noise;

% Detection
bits_rx = rx > 0;
```

```
% BER Calculation
errors = sum(bits ~= bits_rx);
BER = errors/N;

fprintf('SNR = %d dB\n', SNR);
fprintf('Number of bit errors = %d\n', errors);
fprintf('BER = %f\n', BER);

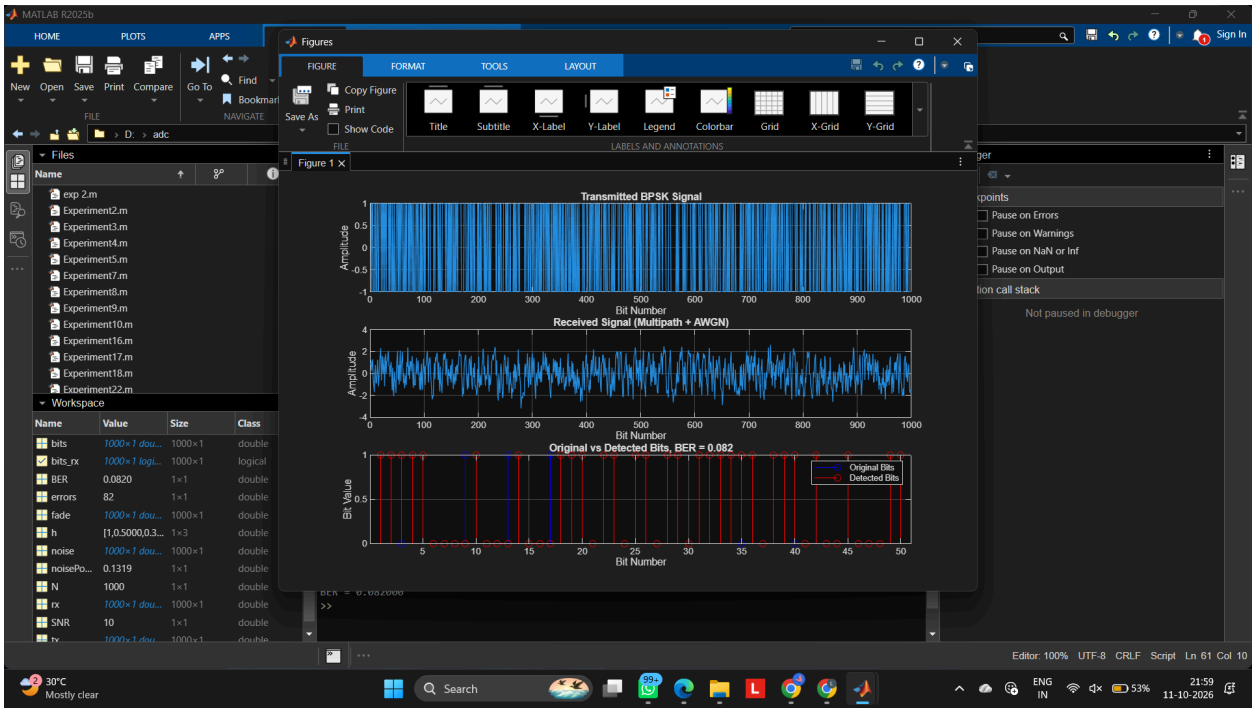
% Plot results
figure;

subplot(3,1,1);
plot(tx);
title('Transmitted BPSK Signal');
xlabel('Bit Number');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(rx);
title('Received Signal (Multipath + AWGN)');
xlabel('Bit Number');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
stem(1:50, bits(1:50), 'b');
hold on;
stem(1:50, bits_rx(1:50), 'r');
title(['Original vs Detected Bits, BER = ' num2str(BER)]);
xlabel('Bit Number');
ylabel('Bit Value');
legend('Original Bits','Detected Bits');
grid on;
hold off;
```

OUTPUT


Figures
Figure 1
Transmitted BPSK Signal
Amplitude
Bit Number
Received Signal (Multipath + AWGN)
Amplitude
Bit Number
Original vs Detected Bits, BER = 0.082
Original Bits
Detected Bits
Bit Value
Bit Number